Joseph Clarke

I am a Senior Scientist at the Met Office Hadley Centre. I am Earth System Scientist with particular interest in nonlinear behaviour in the climate-carbon system. For example, I have analysed Amazon dieback in CMIP6 data, used the complex land-surface model JULES to answer questions about climate-carbon cycle feedbacks, and have designed conceptual models to analyse rate induced tipping. I am interested in topics broadly related to tipping points, such as overshoot, irreversibility, feedbacks and sensitivity. As well as using models, I am interested in empirical approaches to measuring changes in the stability of potential tipping elements using early warning signals. In addition to my background in climate science, I use techniques from dynamical systems, statistical and stochastic dynamics to understand these nonlinear phenomena in the Earth system.

PERSONAL INFORMATION

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EDUCATION

2019–2023 | PhD in Mathematics, **University of Exeter**,
PhD Thesis: 'Tipping Points and Early Warning Signals in the Climate-Carbon System', supervised by Professor Peter Cox and Professor Peter Ashwin.

2015–2019 | MPhys Physics (1st Class), **Brasenose College, Oxford University**,
Specialising in: The Physics of Atmospheres and Oceans and Condensed Matter Physics
Master's Thesis: 'Heat-engine based representation of convection with phase change in exoplanet atmospheres' with Professor Raymond Pierrehumbert.

EMPLOYMENT AND WORK EXPERIENCE

2026– | Senior Scientist, Met Office Hadley Centre
FUTURA project. Researching abrupt shifts in the biosphere

2023–2026 | Postdoctoral Research Associate, University of Exeter
PREDICT (Work Package lead) and WIEMIP projects. With PREDICT I develop new early warning signals for tipping points. With WIEMIP I study warming induced emissions with a focus on overshoot and TCRE.

2019–2023 | Postgraduate Teaching Assistant, University of Exeter
Assisted teaching in the following modules: 'Probability, Statistics and Data', 'Mathematics of Climate Change', 'Ordinary Differential Equations', 'Partial Differential Equations', 'Fluid Dynamics'

2018 | Summer Project with Dr Hannah Christensen, University of Oxford.
Took a dynamical systems approach to constraining climate sensitivity. I worked with Linear Inverse Models, Empirical Orthogonal Functions, autoregressive models, Python,the Fluctuation Dissipation Theorem and undertook analysis of CMIP data

2017 | Summer Intern at the Earth Observation Data Group, University of Oxford.
Developed a website to display observations of atmospheric anomalies in near real time. This involved processing the large amount of data generated and converting it to an understandable form.

POSITIONS OF RESPONSIBILITY

2025–2026 | **XCS Seminar Organiser**
I run the XCS (Exeter Climate Systems) seminars.

2025 | **Global Tipping Points Conference Committee**
Organising the Global Tipping Points 2025 conference, an interdisciplinary conference with over 500 attendees.

2024–2025 | **GSI Seminar Organisation Committee**
We ran the Global Systems Institute seminar series. We invited speakers, organised rooms as well as provided AV support. Events included traditional seminars, assemblies and film screenings.

2024 | **Pint of Science Organiser**
Outreach activity arranging science talks for the public. Involved working with speakers, operating AV equipment and running the events.

RESEARCH EXPERTISE

- Scientific programming using Fortran and Python, including GPU programming using JAX.
- Running and developing land surface models
- Climate Emulators
- Emergent Constraints
- Climate Sensitivity
- Carbon Cycle biogeochemistry
- Tipping Point Early Warning Signals
- Autonomous and nonautonomous dynamical systems techniques (e.g. bifurcation analysis, overshoot, rate induced tipping, linear response theory)

PUBLICATIONS

2026 | Abrams, Jesse F., Chris Huntingford, Sergei Petrovskii, Joshua E. Buxton, Helen Millman, Chris A. Boulton, Robert J. Parker, Cristina Ruiz Villena, Andrew M. Cunliffe, **Joseph Clarke**, Valerio Lucarini, Darren Ghent, Peter M. Cox, and Timothy M. Lenton (2026). "Science Needs for a Terrestrial Biosphere Tipping Point Early Warning System". In: *Reviews of Geophysics* 64.4. DOI: 10.1029/2025RG000900. URL: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2025RG000900.

Abrams, Jesse F., Laura M. Pereira, Viktoria Spaiser, Nico Wunderling, Sebastian Villasante, J. David Tàbara, Reinette Biggs, David I. Armstrong McKay, Caroline Zimm, Manjana Milkoreit, Sara M. Constantino, Ramit Debnath, Vasilis Dakos, Paul D. L. Ritchie, Steven R. Smith, Juan C. Rocha, Sonia Kéfi, Joshua E. Buxton, Chris A. Boulton, Kai Greenlees, Sirkku Juhola, Avit Bhowmik, Lukas P. Fesenfeld, **Joseph Clarke**, James G. Dyke, Laurie Laybourn, Steven J. Lade, Mia Marjanovic-Abrams, Ben Dickenson Bampton, and Timothy M. Lenton (2026). "Integrating Tipping Point Concepts Across Diverse Systems". In: *Communications Sustainability* 1.1, p. 68. ISSN: 3059-4308. DOI: 10.1038/s44458-026-00063-5. URL: https://doi.org/10.1038/s44458-026-00063-5.

Boardman, P. J. L., **J. Clarke**, P. M. Cox, C. Huntingford, C. D. Jones, and M. S. Williamson (2026). "Emergent constraints on climate sensitivity and recent record-breaking warm years". In: *Earth System Dynamics* 17.3, pp. 829–841. DOI: 10.5194/esd-17-829-2026. URL: https://esd.copernicus.org/articles/17/829/2026/.

Clarke, **Joseph**, Chris Huntingford, Paul D L Ritchie, and Peter Cox (May 2026). "Conditions for skilful spatial and temporal tipping point early warning signals". In: *Journal of Physics: Complexity* 7.2, p. 025007. DOI: 10.1088/2632-072X/ae6217. URL: https://doi.org/10.1088/2632-072X/ae6217.

Cox, Peter M., **Joseph J. Clarke**, Simon R. G. Jones, and Rebecca M. Varney (Aug. 2026). "A New Method for Characterising Carbon Cycle Sensitivities in Earth System Models". In: *Tellus*. DOI: 10.16993/tellus.4128. URL: https://doi.org/10.16993/tellus.4128.

Huntingford, C., P. D. L. Ritchie, and **J. Clarke** (2026). "A simple dynamical system for representing climate tipping points with hysteresis". In: *Nonlinear Processes in Geophysics* 33.3, pp. 385–399. DOI: 10.5194/npg-33-385-2026. URL: https://npg.copernicus.org/articles/33/385/2026/.

Ritchie, Paul D L, Norman J Steinert, Jesse F Abrams, Hassan Alkhayuon, Constantin W Arnscheidt, Nils Bochow, Ruth R Chapman, **Joseph Clarke**, Donovan P Dennis, Jonathan F Donges, Bernardo M Flores, Julius Garbe, Annika Högner, Chris Huntingford, Timothy M Lenton, Johannes Lohmann, Kerstin Lux-Gottschalk, Manjana Milkoreit, Tessa Möller, Paul Pearce-Kelly, Laura Pereira, Courtney Quinn, Carl-Friedrich Schleussner, Simone M Stuenzi, Didier Swingedouw, Larissa N Van der Laan, Kirsten Zickfeld, and Nico Wunderling (2026). "The implications of overshooting 1.5 °C on Earth system tipping elements - a review". In: *Environmental Research Letters* 21.4, p. 043001. DOI: 10.1088/1748-9326/ae3cad. URL: https://doi.org/10.1088/1748-9326/ae3cad.

2025 **Clarke**, **J.**, C. Huntingford, P. D. L. Ritchie, R. Varney, M. S. Williamson, and P. Cox (2025). "Conditions for Instability in the Climate-Carbon Cycle System". In: *Earth System Dynamics* 16.6, pp. 2087–2100. DOI: 10.5194/esd-16-2087-2025. URL: https://esd.copernicus.org/articles/16/2087/2025/.
Highlight paper. I found the conditions under which the pre-inustrial climate-carbon cycle system would be linearly unstable.

2024 **Clarke**, **J** (2024). "Tipping Points and Early Warning Signals in the Climate-Carbon System". PhD thesis. University of Exeter. URL: https://hdl.handle.net/10871/134880.
My thesis which examined tipping points in the carbon cycle and investigated early warning signals.

Huntingford, Chris, Peter M Cox, Paul DL Ritchie, **Joseph J Clarke**, Isobel M Parry, and Mark S Williamson (2024). "Acceleration of daily land temperature extremes and correlations with surface energy fluxes". In: *npj Climate and Atmospheric Science* 7.1, p. 84. URL: https://doi.org/10.1038/s41612-024-00626-0.

2023 **Clarke**, **Joseph J**, Chris Huntingford, Paul DL Ritchie, and Peter M Cox (2023). "Seeking more robust early warning signals for climate tipping points: the ratio of spectra method (ROSA)". In: *Environmental Research Letters* 18.3, p. 035006. URL: https://doi.org/10.1088/1748-9326/acbc8d.
Conventional early warning signals make assumptions about the noise the system experiences, which can make them ineffective when those assumptions are not satisfied. I introduced a new early warning signal which still works well for more general, time correlated noise. I applied this to Amazon dieback in CMIP6.

2022 Huntingford, Chris, Peter M Cox, Mark S Williamson, **Joseph J Clarke**, and Paul DL Ritchie (2022). "Emergent constraints for the climate system as effective parameters of bulk differential equations". In: *Earth System Dynamics Discussions* 2022, pp. 1–15. URL: https://doi.org/10.5194/esd-14-433-2023.

Ritchie, Paul DL, Isobel Parry, **Joseph J Clarke**, Chris Huntingford, and Peter M Cox (2022). "Increases in the temperature seasonal cycle indicate long-term drying trends in Amazonia". In: *Communications Earth & Environment* 3.1, pp. 1–8. URL: https://doi.org/10.1038/s43247-022-00528-0.

2021 **Clarke**, **Joseph**, Chris Huntingford, Paul Ritchie, and Peter Cox (2021). "The compost bomb instability in the continuum limit". In: *The European Physical Journal Special Topics*, pp. 1–7. URL: https://doi.org/10.1140/epjs/s11734-021-00013-3.
I derived a conceptual model for the compost bomb, showing that the diffusion of heat does not suppress the compost bomb instability. I demonstrated that compost bombs could be caused by hot summers and not just rapid global warming.

Ritchie, Paul DL, **Joseph J Clarke**, Peter M Cox, and Chris Huntingford (2021). "Overshooting tipping point thresholds in a changing climate". In: *Nature* 592.7855, pp. 517–523. URL: https://doi.org/10.1038/s41586-021-03263-2.

SELECTED TALKS AND RESEARCH SESSIONS

2026 Conditions for Instability in the Climate Carbon Cycle System, **EGU**
Presented in the session The Climate Model Hierarchy: Bridging simulation, understanding and application

Conditions for skilful spatial and temporal tipping point early warning signals, **EXtremes and Rare Events in Climate and related applications (ExREC)**
Invited Speaker

Non-linear ecosystem responses to climate overshoot (JULES), **Warming-Induced Emissions Model Intercomparison Project (WIEMIP): Results and Future Direction**
Takes place July 2026

2025 Tipping Points in the Biosphere, **Global Tipping Points Conference**
I co-organised this research session

2024 Conditions for stability in the Carbon-Climate System, **Dynamics Days Europe 2024**
Invited to present my work on bifurcations in the climate-carbon system

2021 | Early Warning Signals: Beyond White Noise, **Tipping Points Seminar, Potsdam Institute for Climate Impact Research**
Invited to present on early warning signals for systems subject to time correlated noise

2020 | The Compost Bomb in the Continuum Limit, **Emergent Constraints and Tipping Points Workshop**
Took part in workshop developing connections between tipping points and emergent constraints

SUCCESSFUL RESEARCH GRANT APPLICATIONS

2025 | PREDICT
Funded by ESA. Work Package lead

2026 | EO4TIPP
Funded by ESA

RESEARCH SUPERVISION

2024– | Co-supervisor of PhD student Patric Boardman
Thesis title: Emergent Constraints on Future Climate Extremes

REFEREES

PROFESSOR PETER COX: p.m.cox@exeter.ac.uk
PROFESSOR PETER ASHWIN: p.ashwin.exeter.ac.uk